

ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

Implementation of Baseline Model 2 (Regression Approach)

Introduction

Baseline Model 2 explores regression-based approaches to predict sentiment scores. The goal was to evaluate how well numerical prediction models can approximate sentiment values.

Models Used

Two regression models were implemented:

- Linear Regression
- Random Forest Regressor

Model Evaluation

Linear Regression Results:

- Mean Squared Error (MSE): 0.192
- R^2 Score: 0.91

Random Forest Results:

- Mean Squared Error (MSE): 0.225
- R^2 Score: 0.89

Analysis

- Linear Regression performs slightly better than Random Forest
- High R^2 values indicate strong prediction capability
- Models can approximate sentiment trends effectively

Conclusion

Baseline Model 2 demonstrates strong predictive performance in terms of numerical sentiment scoring. However, for practical sentiment classification tasks, classification models are more interpretable and effective.

Overall Insight

While regression models provide good numerical predictions, classification models are more suitable for real-world sentiment analysis tasks. This motivated the transition to deep learning models in the next phase.